

Introduction to Statistical Methods
(S1-24 AIMLCZC418) Assignment 2
AIML Section – 6

Each question carries 2.5 Marks (2.5 x 4 = 10 Marks)

Duration: 15th February 2025 – 2nd March 2025

1) Submissions are individual

2) Solve these on paper, scan, and upload

3) Plagiarism results in zero marks

4) Write your name, BITS ID and Section on each page

1. A textile manufacturer is investigating the effect of two factors on the tensile strength of a synthetic fiber: the type of fiber (Factor A: Nylon, Polyester) and the production machine (Factor B: Machine 1, Machine 2, Machine 3). Two samples were tested for each fiber-machine combination. Perform a two-way ANOVA to assess the effects of these factors on tensile strength at 5% level of significance

Fibre Type	Machine 1	Machine 2	Machine 3
Nylon	35, 37	40, 42	38, 39
Polyester	45, 46	48, 50	47, 49

2. A study was conducted to examine the relationship between the hours of study per week and the exam scores of students in an engineering course. Data were collected from 15 students as shown below:

Study Hours (X)	5	10	15	7	12	20	18	25	30	22	16	8	14	6	28
Exam Score (Y)	55	68	74	60	72	88	85	90	95	87	79	62	75	58	92

Compute the correlation coefficient and interpret its value.

Determine if the correlation is significant at $\alpha = 0.01$.

3. In a study, the waiting times of customers at a bank were recorded. A random sample of 30 customers showed an average waiting time of 5.2 minutes with a standard deviation of 1.8 minutes. Estimate the population mean waiting time with a 90% confidence interval. Also, estimate the population variance with a 95% confidence interval.

4. A civil engineer wants to model the relationship between **traffic volume** (number of cars per hour) and **road surface temperature** (°C). The goal is to determine whether traffic volume influences road surface temperature. Data were collected on **12 different days**, recording the number of cars passing per hour and the corresponding road surface temperature. The data are given below:

Day	Traffic Volume (X) (cars/hour)	Road Surface Temperature (Y) (°C)
1	320	27
2	450	31
3	600	35
4	280	26
5	510	33
6	700	38
7	620	36
8	900	42
9	750	39
10	400	30
11	530	34
12	810	40

Fit a simple linear regression model, check the residuals for normality, and calculate the coefficient of determination.
